

SUYOG HANMANT BARGULE

AI Engineer (Agentic AI & Computer Vision)

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📍 Pune Maharashtra, India



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<https://suyogbargule.github.io/portfolio>

Profile Summary

AI & Computer Vision Engineer with 4+ years of experience in building real-time, production-grade AI systems across surveillance, traffic monitoring, assistive technology, and generative AI. Specialized in computer vision optimization, agentic AI, RAG pipelines, and scalable model deployment. Proven impact with up to 81% accuracy improvement and 80% reduction in manual effort through automation.

Work Experience

Simplifi Dentistry Solutions India Pvt Ltd , Remote– AI Engineer

📅 Oct 2025-Jan 2026

- Automated patient data extraction from insurance portals using Playwright-based web scraping and converted it into structured insurance form data.

Futops Technologies, Pune – AI & Computer Vision Engineer

📅 Oct 2022-Aug 2025

- Developed an agentic AI lead generation system using n8n, Apollo AI, Afify Scraper, and LinkedIn/company data → automated lead identification, analysis, and qualification, reducing manual prospecting by 80% and improving conversion efficiency.
- Built a local LLM-powered chatbot using LangChain and DeepSeek (14B), integrated via Ollama with CUDA → deployed on company website for low-latency, private inference.
- Designed a computer vision traffic surveillance system to detect helmet violations, triple riding, and red-light jumping → improved detection accuracy by 81%. Introduced best-frame snapshot selection, boosting object detection from 40% → 75% (SPOT Award).
- Developed a real-time assistive system for visually impaired users using AWS cloud services → achieved 82% accuracy with a scalable, cloud-native architecture.
- Optimized a face recognition pipeline → 95% recognition, 90% detection accuracy, including age, gender, and mask detection; inference latency 30–35 ms, suitable for edge AI deployment.
- Implemented panoramic 360° CCTV imaging and low-light image stitching using OpenCV and GStreamer → improved surveillance coverage and image clarity (SPOT Award).

AI Engineer Intern (Research & Development) – AI4SEE Pvt. Ltd., Bangalore

📅 Feb 2021-Sep 2022

- Built a Flutter-based rope jump counter using a TensorFlow Lite model for real-time analysis.
- Developed a mobile app for direct farmer-to-retail sales; boosted sales by 20% with a regression model predicting prices five days in advance.

Skills

- Programming Languages:** Python, C++, javascript, CUDA
- Machine Learning & Deep Learning:** Machine Learning Engineer skills, Deep Learning, Supervised & Unsupervised Learning, CNNs, GANs, Vision Transformers, NLP, Multi-modal AI
- GenAI, LLMs & Agentic AI:** Large Language Models (LLMs), LLaMA, DeepSeek, Hugging Face, LangChain, RAG, Agentic AI, LangGraph, LangSmith, LangFlow, Prompt Engineering, LoRA Fine-tuning, Stable Diffusion, Synthetic Data Generation
- Computer Vision:** YOLO, Object Detection, Image Segmentation, Image Processing, Performance Optimization, Face Recognition, Low-Light Imaging, Panoramic Imaging, OCR.
- Libraries & Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, ONNX, OpenVINO, CUDA, NumPy, Pandas
- MLOps, CI/CD & Model Deployment:** MLOps, Model Deployment, Model Monitoring, CI/CD Pipelines, Data Pipelines, ONNX Runtime, TensorRT, FastAPI, REST Full APIs, Docker, Kubernetes,
- Cloud & Infrastructure:** AWS (Rekognition, S3, EC2, SageMaker), Azure, Linux, Kafka, ZeroMQ, GStreamer, OpenGL, Playwright
- Databases & Vector Stores:** SQL, PostgreSQL, MongoDB, Qdrant, Pinecone, FAISS
- Soft Skills:** Teamwork & Collaboration, Adaptability, Analytical Thinking, Problem-Solving, Effective Communication

Languages

English, Hindi, Marathi

Education

Master of Computer Application

Shivaji University Computer Science Department | Graduation Year: 2019

Certifications

Data Science and Machine Learning, ExcelR Solutions, Jan 2019 – May 2019

PyTorch for Deep Learning Bootcamp, Udemy – Andrei Neagoie & Daniel Bourke, Jun 2024 – May 2025